



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

A Cybersecurity Incident Reporting and Risk Assessment System for Detecting, and Prioritizing Security Threats in an Organization.”

CSA0203-C PROGRAMMING

Guided by:

Dr . K. Saravanan

Presented By:

N. Sanjana(192412478)

Janaki(192412409))

Syed Nazma(192425193)



Engineer to Excel



INTRODUCTION

1. Cybersecurity attacks are increasing rapidly.
2. Security incidents need to be reported quickly.
3. Manual reporting can cause delays and errors.
4. Our system organizes incident reporting and risk assessment.
5. It helps improve incident response and management.





Engineer to Excel

PROBLEM STATEMENT



Existing problems:

1. Delayed incident reporting
2. Unorganized incident records
3. Difficulty in tracking incidents
4. Manual risk classification
5. Difficulty handling multiple incidents.





ABSTRACT



1. The system helps organizations report and manage cybersecurity incidents.
2. It detects threats such as phishing, malware, unauthorized access, and data breaches.
3. Each incident is analyzed based on severity, impact, and likelihood.
4. The system prioritizes threats as Low, Medium, High, or Critical.
5. It helps security teams respond quickly, reduce risks, and improve organizational cybersecurity.



EXISTING SYSTEMS



1. **Manual Incident Reporting:** Security incidents are often reported through emails, phone calls, or manual forms.
2. **Separate Security Tools:** Different tools are used for threat detection, monitoring, and incident management.
3. **Delayed Response:** Manual analysis can increase the time required to identify and respond to security threats.
4. **Limited Risk Prioritization:** Existing systems may not effectively prioritize incidents based on severity and impact.

PROPOSED SYSTEM

1. User Authentication

1. Secure login for users and administrators.
2. Different access levels based on user role.

2. Incident Reporting

1. Users can report security incidents by entering the incident type, description, affected system, likelihood, and impact.

3. Automatic Risk Assessment

1. The system calculates the risk score automatically using:
Risk Score = Likelihood × Impact
2. Incidents are classified as **Low, Medium, High, or Critical**.

4. Incident Database

1. All reported incidents are stored in a centralized SQLite database for easy management and tracking.

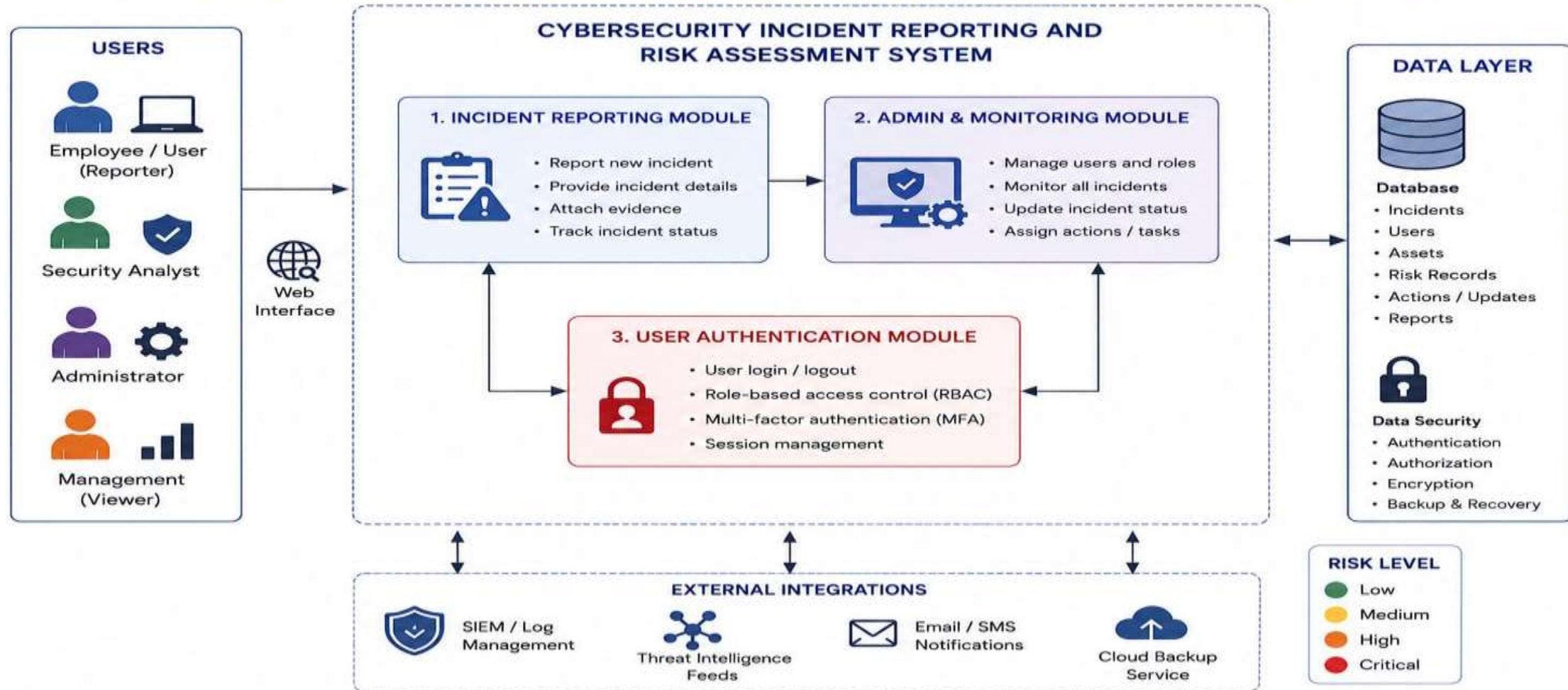


Engineer to Excel



System Architecture

SYSTEM ARCHITECTURE CYBERSECURITY INCIDENT REPORTING AND RISK ASSESSMENT SYSTEM





Engineer to Excel

Module 1 :User Authentication Module

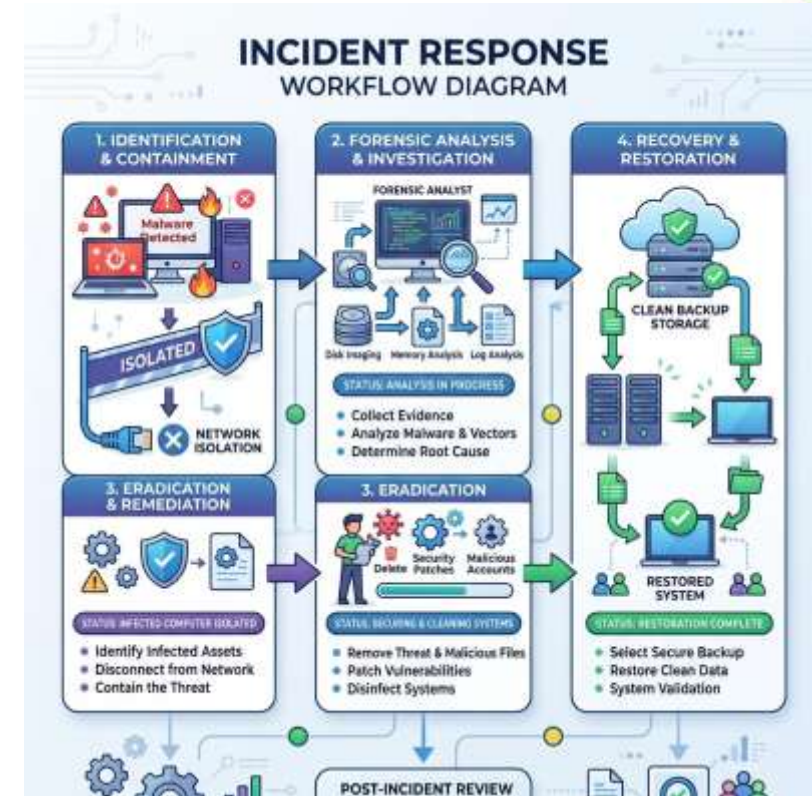


User Authentication Module

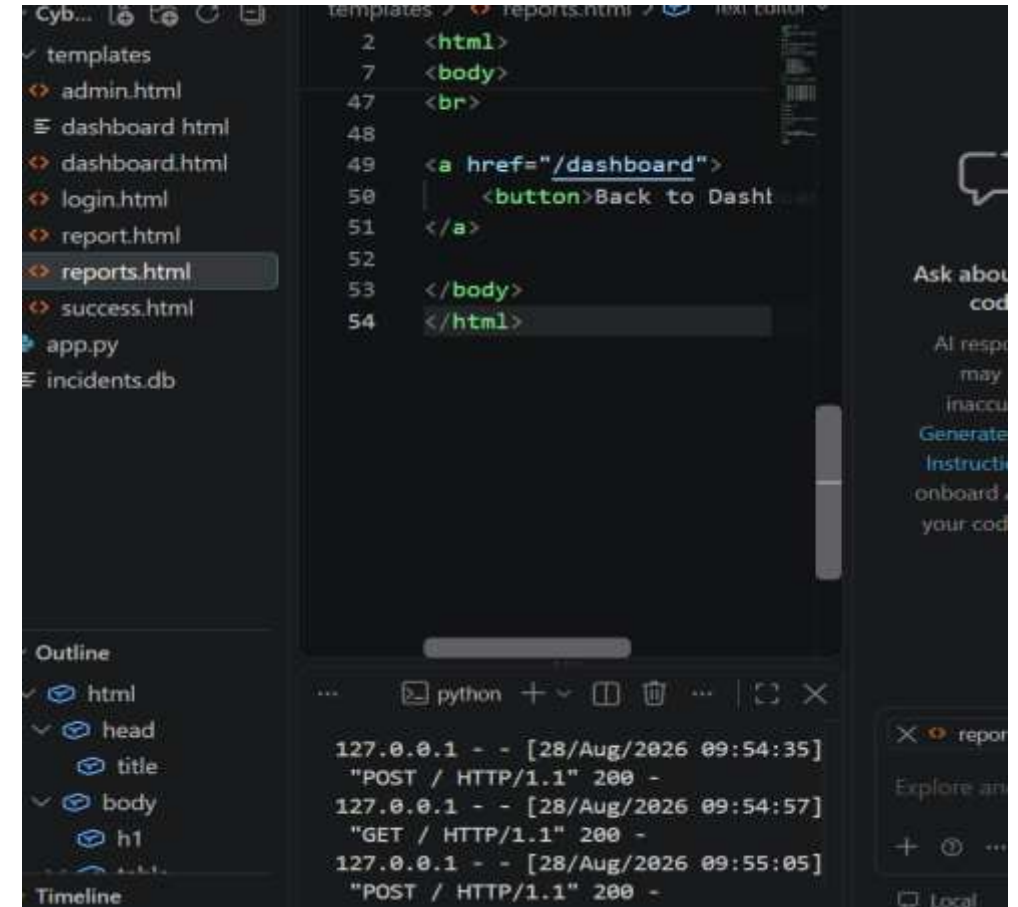
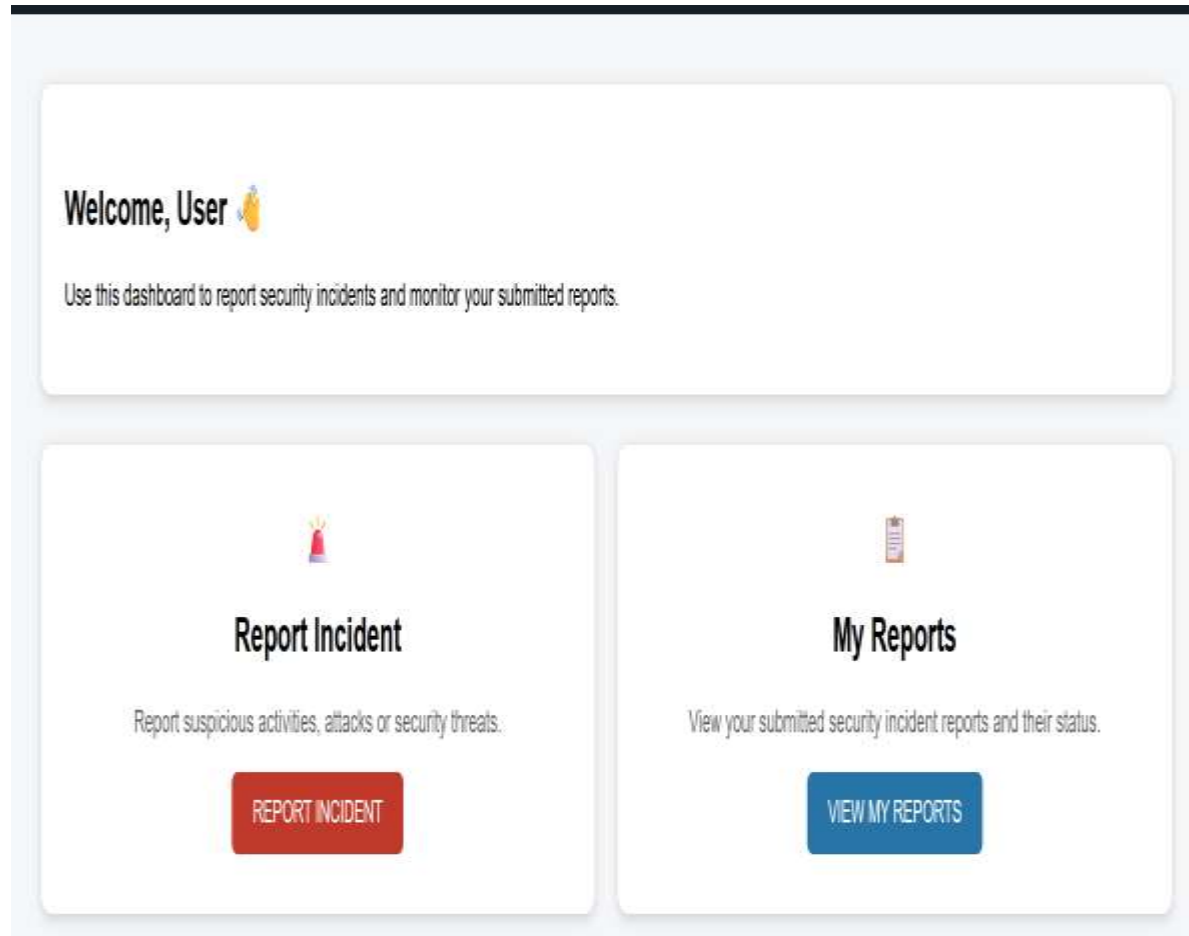
1. Identifies and authenticates users.
2. Allows users to **register**.
3. Provides **secure login**.
4. Verifies user **passwords**.
5. Prevents **unauthorized access**.
6. Allows only authorized users to enter the system.

Main Functions:

Registration → Login → Password Verification → System Access



MODULE 1 OUTPUT





Module 2: Incident Reporting



1. Reports cybersecurity incidents in an organized way
 2. Enters important incident details
 3. Records the reported incident
 4. Uploads supporting evidence
 5. Tracks the status of the reported incident
 6. Helps the administrator investigate and manage the incident
- **Main Functions**
 - **Enter Details → Submit Incident → Upload Evidence → Track Incident**



Engineer to Excel

Module 2: Incident Reporting OUTPUT



Incident Information

Incident Title

Example: Phishing Attack

Incident Type

Select Incident Type

Description

Describe the security incident...

Please fill in this field.

Affected System

Example: Employee PC

Likelihood (1 - 5)

1 - Very Low

Impact (1 - 5)

1 - Very Low

```
templates > success.html > Text Editor
2  <html>
7  <body>

18
19  <a href="/report">
20  |   <button>Report Another
21  </a>
22
23  <br><br>
24
25  <a href="/">
26  |   <button>Logout</button>
27  </a>
28
29  </body>
30  </html>
```

```
127.0.0.1 - - [28/Aug/2026 09:54:35]
"POST / HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 09:54:57]
"GET / HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 09:55:05]
```


Module 3 : Admin Dashboard

1. Provides a central place to manage incidents
2. Displays all reported incidents
3. Allows administrators to review incidents
4. Shows incident risk levels
5. Updates incident status
6. Helps manage and resolve incidents
7. Generates reports for analysis



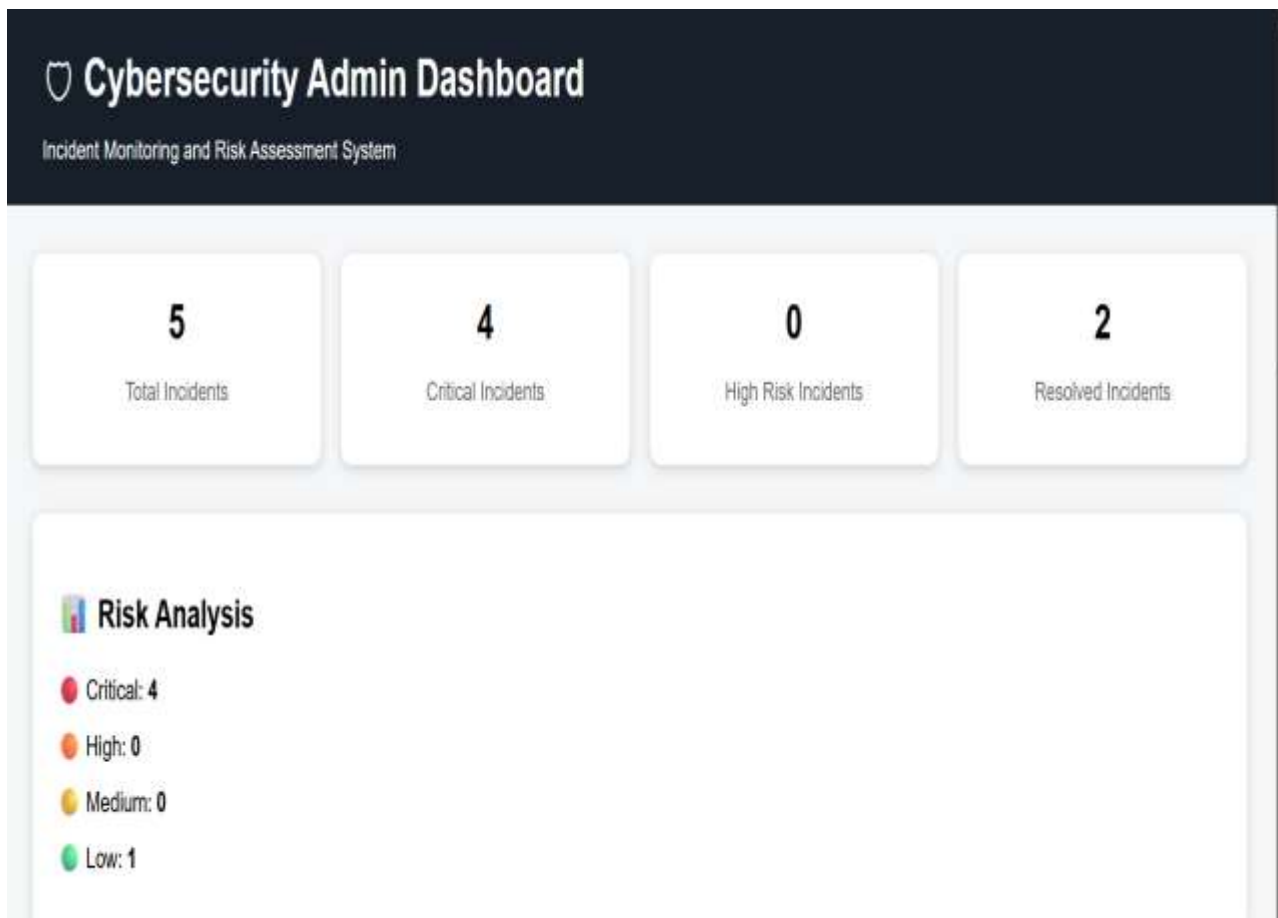
Main Functions

- **View Incidents → Review → Update Status → Resolve → Generate Report**



Engineer to Excel

Module 3 : Admin Dashboard Output



```
<code>admin.html X dashboard Text Editor
templates > admin.html
2 <html>
222 <body>
237 <div class="container">
399 <div class="table-box">
598
599     {% endif %}
600
601 </div>
602
603
604
605
606 <!-- LOGOUT -->
607
608 <a href="/">
609
610     <button class="logout">
611         LOGOUT
612     </button>
613</code>
```

```
<code>127.0.0.1 - - [28/Aug/2026 09:54:35]
"POST / HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 09:54:57]
"GET / HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 09:55:05]
"POST / HTTP/1.1" 200 -</code>
```



Engineer to Excel

FUNCTIONALITIES



1. **User Authentication** – Provides secure login for users and administrators.
2. **Incident Reporting** – Allows users to report cybersecurity incidents with relevant details.
3. **Risk Assessment** – Calculates the risk score based on likelihood and impact.
4. **Risk Classification** – Categorizes incidents as Low, Medium, High, or Critical.
5. **Incident Management** – Stores and manages reported incidents in a centralized database.
6. **Incident Tracking** – Allows users and administrators to monitor incident status.
7. **Admin Dashboard** – Provides administrators with an overview of all reported incidents.

Advanced Applications

- 1. AI/ML-Based Risk Prediction**
Predict incident severity and risk automatically.
- 2. Real-Time Security Alerts**
Immediately notify administrators about critical incidents.
- 3. Cloud-Based Deployment**
Enable secure access and centralized incident management.
- 4. Advanced Analytics**
Analyze incident patterns and identify recurring threats.
- 5. SIEM Integration**
Connect the system with security monitoring platforms.



CONCLUSION

- the system provides a structured platform for cybersecurity incident reporting.
- It helps in identifying and classifying security risks.
- Administrators can monitor, track and manage incidents efficiently.
- Secure incident information management improves the overall incident response process.
- Report generation supports better analysis and decision-making.
- The system can be further enhanced using AI/ML and cloud technologies.



Engineer to Excel



References

1. **Halawi Ghoson, N., & Benfriha, K. (2026).**
A novel dynamic risk assessment tool for evaluating cybersecurity risks in the digital industry.
Scientific Reports.
DOI: 10.1038/s41598-026-59483-x
2. **Agathangelos-Ioannou, G., & Tsohou, A. (2026).**
Methodologies for information security risk assessment: Shortcomings for analysing artificial intelligence systems. **International Journal of Information Security, 25, 98.**
DOI: 10.1007/s10207-026-01271-9
3. **Kincl, J., Adam, M. T. P., & Pavleska, T. (2026).**
Human-AI Integration in Cybersecurity: An Industry-Aligned Perspective on Incident Management. **International Journal of Information Security, 25, 103.**
DOI: 10.1007/s10207-026-01259-5
4. **Paragioudakis, A., Komninos, N., Smyrlis, M., et al. (2026).**
Risk management for cyber insurance: A survey for data-driven approaches with the use of AI.
International Journal of Information Security, 25, 112.
DOI: 10.1007/s10207-026-01275-5



Thank You